

PROJECT SUMMARY:

PROBLEM: Kenya faces a KSh 465B water funding gap (NAWASIP), receiving under half of the KSh 100B needed yearly for 2030 coverage. With scarce funds, allocation defaults to political visibility rather than measured need, leaving 3.99M citizens facing water disruptions.

USERS: County water/finance depts, National Treasury, WASREB regulator, and WASH donors (KfW, World Bank).

MARKET SIZE: All 47 Kenya counties sitting within the KSh 465B national water capital gap, covering 55.7M citizens by 2026.

VALUE PROP: A population-weighted AI urgency score per county. Combines a 32-feature SMOTE LightGBM model (0.9205 ROC-AUC), household vulnerability (WVI), governance benchmark (WGS), and census projections into one transparent ranking with SHAP explainability.

REVENUE MODEL: (1) Free public dashboard for civic transparency; (2) Paid B2G subscription for county depts & WASREB (API, custom reporting); (3) Grant-funded donor pilot packages.

UNFAIR ADVANTAGE: We pitch a live working product, not slides. Trained on 21,347 KHS 2023/24 households and 2019 Census projections into a deployed app: <https://water-futures.vercel.app/>

GENERAL COMMENTS

- Aligned to the theme.
- Uses data and AI to inform water-investment allocation.
- Evidence of private sector and government partnerships.

